UE24CS320A-– Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	01 Aug 2026	**Week Ending Date**	09 Aug 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Explored broad cybersecurity domains and shortlisted incident response, ransomware resilience, and post-incident recovery as possible capstone directions.	01–03 Aug 2026	Initial domain exploration focused on a feasible cybersecurity problem with a clear real-world need.
2	Compared candidate problem areas by relevance, technical feasibility, available evidence, and suitability for a Phase-I proposal.	04–06 Aug 2026	Options were evaluated at a preliminary level; no final problem statement was fixed.
3	Discussed how recovery decisions can remain difficult even after a cyberattack has been detected and contained.	07–09 Aug 2026	This helped narrow the search toward trustworthy recovery rather than building a new attack detector.

In Progress

#	Tasks	Date	Remarks
1	Searching for a specific, researchable problem statement within cybersecurity and AI.	01–09 Aug 2026	Candidate directions remained under consideration.

Plan for Next Week

#	Tasks	Date	Remarks
1	Continue comparing candidate problem statements and identify a focused recovery-related problem.	10–13 Aug 2026	Prioritise scope, feasibility, and a demonstrable research contribution.
2	Collect introductory sources on ransomware recovery and incident-response practices.	10–16 Aug 2026	Use credible standards, reports, and research literature to inform selection.

Dr. Geetha Dayalan

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UE24CS320A-- Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	10 Aug 2026	**Week Ending Date**	16 Aug 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Reviewed the stages of a typical ransomware incident, including compromise, detection, containment, and restoration.	10–12 Aug 2026	Focused on the recovery stage after an incident has been identified.
2	Examined the role of backups and disaster-recovery procedures in restoring affected enterprise services.	12–14 Aug 2026	Identified the risk of assuming that the most recent backup is necessarily clean.
3	Compared possible project directions and refined the search toward selecting trustworthy recovery points.	14–16 Aug 2026	Problem-statement exploration continued; no mentor approval is recorded for this week.

In Progress

#	Tasks	Date	Remarks
1	Investigating whether evidence from logs, attack timelines, and backup integrity can inform recovery decisions.	10–16 Aug 2026	Preliminary concept only; scoring criteria still require research and refinement.
2	Searching for academic and industry sources on backup compromise and safe recovery.	10–16 Aug 2026	Sources are being gathered to establish the problem context.

Plan for Next Week

#	Tasks	Date	Remarks
1	Narrow the candidate problem statement around trustworthiness of recovery points after ransomware.	17–20 Aug 2026	Keep the proposed scope manageable for a capstone prototype.
2	Review relevant standards and papers on incident response, recovery, and security automation.	17–23 Aug 2026	Record the methods, assumptions, and limitations of each source.

Dr. Geetha Dayalan

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UE24CS320A-– Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	17 Aug 2026	**Week Ending Date**	23 Aug 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Refined the problem focus to the difficulty of determining which assets and backups remain trustworthy after an attack.	17–19 Aug 2026	This became the central problem direction under consideration.
2	Reviewed initial material on incident-response guidance, recovery planning, and security orchestration.	19–21 Aug 2026	The review highlighted the reliance on manual investigation and predefined playbooks.
3	Discussed the trade-off between restoring quickly and avoiding reinfection from a compromised recovery point.	21–23 Aug 2026	The problem framing began to connect recovery safety with response time.

In Progress

#	Tasks	Date	Remarks
1	Searching for research papers on AI-assisted incident response and recovery-point selection.	17–23 Aug 2026	Literature collection is preliminary and continues in subsequent weeks.
2	Evaluating the feasibility of an evidence-based trust assessment rather than a new detection engine.	17–23 Aug 2026	The project direction is being scoped to decision support and recovery planning.

Plan for Next Week

#	Tasks	Date	Remarks
1	Develop a clearer problem statement and identify the specific gap in existing recovery workflows.	24–27 Aug 2026	Distinguish the proposed contribution from general incident detection.
2	Continue collecting papers and standards that can support the background and motivation.	24–30 Aug 2026	Capture bibliographic details and key findings for later comparison.

Dr. Geetha Dayalan

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UE24CS320A-– Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	24 Aug 2026	**Week Ending Date**	30 Aug 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Outlined the need to assess recovery-point trust using evidence such as attack timelines, alerts, and integrity information.	24–26 Aug 2026	Conceptual direction; no implementation claim is made.
2	Identified the importance of distinguishing trusted, uncertain, and untrusted recovery points.	26–28 Aug 2026	Uncertainty should be represented explicitly when evidence is incomplete.
3	Drafted the initial rationale for ranking recovery options while considering recovery safety and data loss.	28–30 Aug 2026	This provided a basis for discussing a measurable project objective.

In Progress

#	Tasks	Date	Remarks
1	Refining the candidate problem statement and checking its feasibility against the Phase-I schedule.	24–30 Aug 2026	Problem selection and wording remain subject to mentor discussion.
2	Building a preliminary reading list on threat intelligence, backup security, and AI agents.	24–30 Aug 2026	The reading list will support a more structured literature survey.

Plan for Next Week

#	Tasks	Date	Remarks
1	Compare existing approaches and record limitations relevant to trustworthy recovery.	31 Aug–03 Sep 2026	Focus on what existing methods do not directly address.
2	Draft a concise problem statement, objectives, and tentative scope for mentor discussion.	31 Aug–06 Sep 2026	Keep claims proportional to the planned Phase-I work.

Dr. Geetha Dayalan

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UE24CS320A-- Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	31 Aug 2026	**Week Ending Date**	06 Sep 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Reviewed existing recovery workflows and noted their dependence on manual judgement or predefined playbooks.	31 Aug–02 Sep 2026	This informed the motivation for evidence-driven recovery decisions.
2	Organised the problem around assessing backup trust, ranking recovery points, and planning restoration safely.	02–04 Sep 2026	The proposed work remained a conceptual framework at this stage.
3	Considered constraints including incomplete evidence, changing threat information, and service dependencies.	04–06 Sep 2026	These constraints were carried forward into the initial scope.

In Progress

#	Tasks	Date	Remarks
1	Reviewing papers and guidance on AI-assisted response, security orchestration, and backup recovery.	31 Aug–06 Sep 2026	Detailed paper-by-paper synthesis is still ongoing.
2	Refining the wording and boundaries of the proposed problem statement.	31 Aug–06 Sep 2026	The mentor approval date had not yet occurred.

Plan for Next Week

#	Tasks	Date	Remarks
1	Prepare a structured comparison of existing work and the proposed research gap.	07–10 Sep 2026	Separate established capabilities from the missing trust-and-recovery link.
2	Draft the proposed workflow and preliminary evaluation criteria.	07–13 Sep 2026	Consider recovery time, correct backup selection, and unsafe recovery outcomes.

Dr. Geetha Dayalan

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UE24CS320A-– Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	07 Sep 2026	**Week Ending Date**	13 Sep 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Organised background research into incident-response standards, conventional recovery workflows, and AI-agent approaches.	07–09 Sep 2026	The categories helped structure the background-work section.
2	Identified preliminary gaps around measurable recovery-point trust, evidence updates, and safety controls.	09–11 Sep 2026	These were treated as research hypotheses to validate against sources.
3	Outlined a conceptual workflow linking incident evidence, trust assessment, policy checks, and recovery planning.	11–13 Sep 2026	The workflow was intended for the proposal, not yet implemented.

In Progress

#	Tasks	Date	Remarks
1	Verifying research-paper findings and references before using them in the presentation.	07–13 Sep 2026	Statistics and claims require traceable sources.
2	Developing a preliminary explanation of the project contribution and expected evaluation measures.	07–13 Sep 2026	The proposed evaluation remains to be formalised.

Plan for Next Week

#	Tasks	Date	Remarks
1	Refine the project scope, feasibility, assumptions, and out-of-scope items.	14–17 Sep 2026	Emphasise controlled-lab evaluation and human approval for high-impact actions.
2	Continue the literature survey and map sources to each proposed research gap.	14–20 Sep 2026	Maintain a source-to-claim record for the review presentation.

Dr. Geetha Dayalan

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UE24CS320A-- Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	14 Sep 2026	**Week Ending Date**	20 Sep 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Developed the preliminary workflow for understanding an attack, scoring recovery-point trust, applying policy, and planning restoration.	14–16 Sep 2026	This workflow was documented as the proposed approach.
2	Considered a trust score and three trust states to communicate whether a recovery point can be used, reviewed, or excluded.	16–18 Sep 2026	Thresholds and scoring rules remain to be justified and validated.
3	Outlined safety measures including policy gates, human approval, and post-recovery verification.	18–20 Sep 2026	The proposal keeps consequential actions under human oversight.

In Progress

#	Tasks	Date	Remarks
1	Analysing service dependencies and the need to restore critical services in a safe order.	14–20 Sep 2026	Dependency handling is part of the conceptual design.
2	Continuing detailed literature review on agentic AI, threat intelligence, backup security, and recovery metrics.	14–20 Sep 2026	Research synthesis remains ongoing.

Plan for Next Week

#	Tasks	Date	Remarks
1	Consolidate the research gap, proposed approach, scope, feasibility, and challenge-mitigation sections.	21–24 Sep 2026	Align each section with the stated problem and planned deliverables.
2	Prepare an illustrative ransomware-recovery scenario to explain the workflow.	21–27 Sep 2026	Use it to clarify inputs, trust states, policy checks, and outputs.

Dr. Geetha Dayalan

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UE24CS320A-- Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	21 Sep 2026	**Week Ending Date**	27 Sep 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Organised the motivation and problem framing around the risk of restoring from a compromised or contaminated backup.	21–23 Sep 2026	The rationale was linked to practical ransomware-recovery challenges.
2	Prepared preliminary scope and feasibility material, including a controlled-lab setting and human approval for high-impact steps.	23–25 Sep 2026	Live production deployment and new detection-engine development were kept outside the proposed scope.
3	Structured the proposed contribution as evidence-based trust assessment, policy-controlled planning, and dependency-aware restoration.	25–27 Sep 2026	The wording was refined for the Review 1 presentation.

In Progress

#	Tasks	Date	Remarks
1	Checking citations and research claims used in the background and motivation sections.	21–27 Sep 2026	Reference verification is still required before final submission.
2	Refining the illustrative scenario and the criteria intended for later evaluation.	21–27 Sep 2026	No experimental results are claimed at this stage.

Plan for Next Week

#	Tasks	Date	Remarks
1	Assemble the first complete version of the Capstone Review 1 presentation.	28 Sep–01 Oct 2026	Cover the problem, background, research gap, approach, scope, feasibility, and references.
2	Review slide consistency and ensure the proposed contribution is clearly distinguished from existing work.	28 Sep–04 Oct 2026	Check terminology and avoid overstating completed work.

Dr. Geetha Dayalan

Digital Signature of Project Guide With Name

UE24CS320A-- Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	28 Sep 2026	**Week Ending Date**	04 Oct 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Prepared and refined the Review 1 presentation covering the problem statement, background work, research gaps, proposed approach, scope, and feasibility.	28 Sep–02 Oct 2026	The presentation was assembled around the Phase-I proposal deliverables.
2	Added an illustrative ransomware-recovery scenario showing how evidence can inform recovery-point trust and restoration order.	29 Sep–03 Oct 2026	The scenario is explanatory and does not represent an implemented system.
3	Organised references and aligned the proposed workflow with the identified research gaps.	02–04 Oct 2026	Further citation and wording checks remained necessary.

In Progress

#	Tasks	Date	Remarks
1	Conducting a final content review of research-paper summaries, citations, and technical terminology.	28 Sep–04 Oct 2026	The literature survey is not considered complete.
2	Preparing to explain limitations, assumptions, and safety controls during the review.	28 Sep–04 Oct 2026	Focus on feasibility and the distinction between a proposed design and implementation.

Plan for Next Week

#	Tasks	Date	Remarks
1	Incorporate final mentor/team feedback and rehearse the Review 1 presentation.	05–08 Oct 2026	Prioritise clarity of the problem, research gap, and contribution.
2	Complete a consistency check across the abstract, problem statement, proposed approach, and timeline.	05–11 Oct 2026	Ensure all sections use consistent terminology.

Dr. Geetha Dayalan

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UE24CS320A-– Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	05 Oct 2026	**Week Ending Date**	11 Oct 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Consolidated the Review 1 deliverables: problem definition, background research, preliminary research gaps, conceptual workflow, scope, and feasibility.	05–08 Oct 2026	These are proposal-stage deliverables; implementation and experimental validation are future work.
2	Revised the presentation to clarify trust states, policy checks, human approval, and post-recovery verification.	07–09 Oct 2026	The changes strengthened the explanation of the intended safety controls.
3	Reviewed the illustrative scenario, references, and slide sequence in preparation for the review.	09–11 Oct 2026	Presentation readiness was prioritised for the scheduled review.

In Progress

#	Tasks	Date	Remarks
1	Continuing detailed literature survey and paper screening to substantiate the research gaps.	05–11 Oct 2026	A more systematic comparison and synthesis are still required.
2	Preparing notes for questions about feasibility, evaluation metrics, and limitations.	05–11 Oct 2026	Metrics are proposed for later evaluation, not reported as measured results.

Plan for Next Week

#	Tasks	Date	Remarks
1	Present the proposal at Capstone Review 1 and record feedback for follow-up.	12–13 Oct 2026	Review took place on 13 October 2026.
2	Organise post-review action items and identify sections requiring clarification or further evidence.	14–18 Oct 2026	Use feedback to guide the next phase of research and planning.

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UE24CS320A-– Capstone Project Phase – I - Weekly Status Report

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery		
Batch No	221		
Week Starting Date	12 Oct 2026	**Week Ending Date**	18 Oct 2026
Name	Abhirama H Y Adarsh Prakash Dhruva G Hegde Aditya B Jambagi	**SRN**	PES2UG24CS023 PES2UG24CS027 PES2UG24CS671 PES2UG24CS030

Activities Completed

#	Tasks	Date	Remarks
1	Presented the capstone proposal during Capstone Review 1 on 13 October 2026.	13 Oct 2026	The review covered the proposed problem, research motivation, gaps, approach, scope, and feasibility.
2	Reviewed the presentation after the review and organised follow-up points for continued refinement.	14–16 Oct 2026	Only proposal-stage work is recorded; no implementation results are claimed.
3	Updated the immediate work priorities to focus on stronger literature synthesis and clearer evidence for the research gap.	16–18 Oct 2026	These activities prepare the project for the next stage.

In Progress

#	Tasks	Date	Remarks
1	Conducting a detailed literature survey of research papers on agentic AI for incident response, ransomware recovery, backup integrity, and recovery-point selection.	12–18 Oct 2026	Compare objectives, methods, datasets, evaluation metrics, limitations, and relevance to the proposed project.
2	Verifying references and mapping each paper to the identified gaps: trust scoring, changing evidence, safe action gating, and service dependencies.	12–18 Oct 2026	The gap analysis is being refined; problem-statement approval is not recorded as completed in this reporting period.
3	Refining the proposed Trust-Conditioned Recovery Planning concept and the intended evaluation criteria.	12–18 Oct 2026	Design-level refinement only; implementation and validation remain future work.

Plan for Next Week

#	Tasks	Date	Remarks
1	Continue screening and analysing relevant research papers and prepare a structured literature-survey matrix.	19–25 Oct 2026	Record each paper's method, contribution, limitation, and relationship to the research gap.
2	Refine the problem statement, objectives, and proposed methodology using the literature findings and mentor feedback.	19–25 Oct 2026	Seek/record formal approval when it occurs; do not pre-date approval.
3	Develop a more detailed system workflow and define candidate trust-score factors, policy rules, and evaluation scenarios.	19–25 Oct 2026	Keep design decisions evidence-based and feasible for a controlled prototype.

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